

Fig. 1: Single-core CPU processing score for high-end SoCs of key players

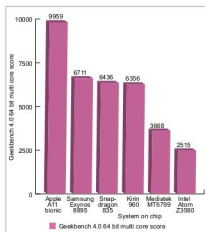


Fig. 2: Multi-core CPU processing score for high-end SoCs of key players

The extended memory—LPDDR2 to LPDDR4—also adds value to the product (smartphone). ARM's big.Little architecture helps smartphone chipsets with battery-saving and powerful cores such as HiSilicon Kirin 960/950/920 and Samsung Exynos 8895.

For instance, when a smartphone user plays a game that requires a high clock speed, processing speed and memory, cores in the processor are divided into big and little cores. Now, load is assigned to each of the cores simultaneously, avoiding situations where the smartphone hangs or its battery dies.

Tables I and II show the number of cores, memory capacity and fabrication technology used in SoCs from various manufacturers. FinFET (Fin field-effect transistors) technology creates thin fin-like structures, which reduces energy consumption.

Chipsets listed in Table I are fabricated on 10nm scale and use advanced memory type (LPDDR4) and features that make them ideal for high-end smartphone devices.

SoC classification as per specifications helps users to understand the SoCs' significance and parameters. Moreover, smartphone manufacturers can compare the performance of the latest and running SoC models and decide to invest accordingly.

Other than FinFET, new technology upgrades include Vulkan API and VolTE technology. Some manufacturers have also started integrating artificial intelligence (AI) features into their chipsets. Kirin 970 chipset is an example.

Upgrades in SoC technology

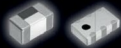
As consumers demand highly efficient devices, "there are a few technology upgrades in chip fabrication to

Attracting Tomorrow **TDK**



TDK Technology Linking the world.

Thin-film and multilayer RF chip components



Miniaturized pressure sensors for highest precision

Power management modules with high efficiency based on SESUB embedding



Advanced lens actuators contribute to higher image quality

<https://product.tdk.com>